A First Course in Abstract Algebra

Section 16: Group Action on a Set

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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16 Group Action on a Set

For a group action, gx denotes the result of applying g to x; it need not be multiplication in a group containing x. The axioms are $ex = x$ and $(gh)x = g(hx)$. The orbit Gx consists of all possible images of x, the stabilizer G_x consists of the group elements fixing x, and X_g is the set of points fixed by one specified g. Keep these three sets distinct. In the classification exercises, a transitive set is understood to be nonempty, as in the book's examples. If the empty set is admitted, the quantified transitivity condition holds vacuously there and adds one empty case.

Exercise 16.1: All fixed sets in the square action

Plan. In each row of Table 16.10, retain exactly the column labels that the row leaves unchanged. Write X for the full seventeen-element set.

$$X_{\rho_0} = X, \qquad X_{\rho_1} = \{C\}, \qquad X_{\rho_3} = \{C\},$$
$$X_{\rho_2} = \{m_1, m_2, d_1, d_2, C\},$$
$$X_{\mu_1} = \{s_1, s_3, m_1, m_2, C, P_1, P_3\},$$
$$X_{\mu_2} = \{s_2, s_4, m_1, m_2, C, P_2, P_4\},$$
$$X_{\delta_1} = \{2, 4, d_1, d_2, C\},$$
$$X_{\delta_2} = \{1, 3, d_1, d_2, C\}.$$

For example, a half-turn exchanges opposite vertices, sides, and midpoints, but leaves each axis and each diagonal as a set. An object in X can be a whole line or side, so "fixed" does not require each point of that object to be fixed individually.

Exercise 16.2: Every stabilizer in the square action

Plan. Read down a column and collect the row labels that reproduce its heading. Let $e = \rho_0$. The answers can be grouped by identical stabilizers:

x	G_x
$1, 3$	$\{e, \delta_2\}$
$2, 4$	$\{e, \delta_1\}$
s_1, s_3, P_1, P_3	$\{e, \mu_1\}$
s_2, s_4, P_2, P_4	$\{e, \mu_2\}$
m_1, m_2	$\{e, \rho_2, \mu_1, \mu_2\}$
d_1, d_2	$\{e, \rho_2, \delta_1, \delta_2\}$
C	D_4

For example, vertex 1 is fixed by the identity and by reflection in its own diagonal d_2, and by no other square symmetry. The orbit–stabilizer check gives orbit size $8/2 = 4$, as expected for the vertices.

Exercise 16.3: The six orbits of the square action

Starting from a vertex reaches every vertex and no other kind of object. The same observation for each kind of object gives

$$\{1, 2, 3, 4\}, \quad \{s_1, s_2, s_3, s_4\}, \quad \{m_1, m_2\}, \quad \{d_1, d_2\}, \quad \{C\}, \quad \{P_1, P_2, P_3, P_4\}.$$

They are disjoint, their sizes sum to 17, and each is closed under all square symmetries. This verifies that the list exhausts X.

Exercise 16.4: Correcting faithful action

The quantifier "for every x" is essential. An action is faithful if

$$(gx = x \text{ for every } x \in X) \implies g = e.$$

Equivalently, the homomorphism $G \to S_X$ determined by the action is injective. Requiring this implication for a single fixed point would mean that all point stabilizers are trivial, which is the stronger condition called a free action. A reflection of a square fixes some vertices without acting as the identity on all vertices.

Exercise 16.5: Correcting transitive action

For every $x, y \in X$ there must exist $g \in G$ such that $gx = y$. The group element may depend on both points. One fixed g cannot send a single x to several different outputs. For a nonempty X, an equivalent formulation is: the orbit of any one chosen point is all of X.

Exercise 16.6: Subsets preserved by an action

Conclusion. The sub-G-sets are precisely the unions of whole G-orbits.

If S is a sub-G-set and $s \in S$, closure gives $Gs \subseteq S$. Since $s = es$, every s is in its own orbit, so $S = \bigcup_{s \in S} Gs$. Conversely, if S is a union of orbits and $s \in Gx \subseteq S$, then $s = hx$ for some h. For any g, $gs = (gh)x \in Gx \subseteq S$. The empty union also yields the empty sub-G-set.

Exercise 16.7: Transitivity in terms of orbits

For nonempty X, transitivity is equivalent to having exactly one orbit. If every y equals gx for some g, then $Gx = X$. Conversely, when $Gx = X$, any $y = hx$ and $z = kx$ satisfy $(kh^{-1})y = z$. If empty sets are allowed under the literal quantified definition, the corresponding characterization is "at most one orbit."

Exercise 16.8: True or false: action properties

(a) False. A group can act on any set; a three-element set with the trivial $\mathbb{Z}_2$ action need not come equipped with a group operation.

(b) True. This is the axiom $ex = x$.

(c) False. In the trivial action every g fixes every point; faithfulness is an additional condition.

(d) True. Apply g^{-1} to $gx_1 = gx_2$ and use the action axioms to get $x_1 = x_2$.

(e) False. In the trivial action of a nontrivial group all group elements send a chosen x to the same point.

(f) True. It is preserved by G, and if hx, kx are two points in the orbit then kh^{-1} sends the first to the second.

(g) True. Restrict the action map to $H \times X$; the same two axioms hold.

(h) False. Take $H = \{e\}$ inside a group that acts transitively on more than one point. Its orbits are singletons, while the original orbit is larger.

(i) True. The inverse of the map $x \mapsto gx$ is $x \mapsto g^{-1}x$.

(j) True. Orbit–stabilizer gives $|Gx| = [G : G_x] = |G|/|G_x|$ when G is finite.

Exercise 16.9: Isomorphic orbits and other subactions

(a) The side orbit and midpoint orbit are isomorphic by $s_i \mapsto P_i$. Every square symmetry sends the midpoint of a side to the midpoint of its image side, so this bijection commutes with the action.

(b) The reflection μ_1 fixes two sides, s_1, s_3, but no vertices. An equivariant bijection must carry the fixed points of any g bijectively onto the fixed points of the same g. Thus no such bijection can identify the vertex and side orbits.

(c) No. The question permits sub-D_4-sets that are unions of orbits. For instance $\{s_1, s_2, s_3, s_4, C\}$ and $\{P_1, P_2, P_3, P_4, C\}$ are different isomorphic sub-D_4-sets: extend the bijection in part (a) by $C \mapsto C$. Among individual orbits, the only isomorphic pair is indeed the side and midpoint pair: the other four-point orbit is excluded by part (b), while μ_1 fixes both axes and exchanges the diagonals, distinguishing the two two-point orbits.

Exercise 16.10: Faithfulness of the square action

(a) Yes. A symmetry fixing all seventeen objects fixes all four vertices. Only the identity symmetry does that.

(b) The vertex, side, and midpoint orbits each give faithful actions: inspection of their columns in Table 16.10 shows that no nonidentity row fixes the entire orbit. The axis orbit and diagonal orbit are not faithful, since the nonidentity half-turn ρ_2 fixes both members in either case. The center orbit is not faithful, since every symmetry fixes C.

Exercise 16.11: Faithfulness and distinct permutations

What we must show. Prove that faithfulness says precisely that different group elements induce different maps on X. Suppose first that the action is faithful and $gx = hx$ for every x. Applying h^{-1} gives $(h^{-1}g)x = x$ for all x, so $h^{-1}g = e$ and $g = h$. Conversely, if distinct group elements have distinct actions and $gx = x$ for every x, then g and e induce the same map. Hence $g = e$, proving faithfulness. Equality of actions here means equality on *every* point, not merely agreement at one point.

Exercise 16.12: Pointwise stabilizers of subsets

Let $G_Y = \{g : gy = y \text{ for every } y \in Y\}$. The identity belongs to G_Y. If $g, h \in G_Y$, then for every $y \in Y$, $(gh)y = g(hy) = gy = y$, so $gh \in G_Y$. If $g \in G_Y$, applying g^{-1} to $gy = y$ gives $g^{-1}y = y$, so $g^{-1} \in G_Y$. Thus G_Y is a subgroup. Equivalently $G_Y = \bigcap_{y \in Y} G_y$. When $Y = \varnothing$, all conditions are vacuous and $G_Y = G$.

Exercise 16.13: The real rotation action

(a) For $\theta \in (\mathbb{R}, +)$ use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. We have $R_0 = I$ and the angle-addition identities give $R_{\theta+\phi} = R_\theta R_\phi$. These are the two action axioms.

(b) Rotation preserves distance to the origin. Conversely every point at distance $|P| > 0$ has an angle differing from that of P by some real θ. Hence the orbit is the entire circle centered at the origin through P.

(c) A nonzero point returns to itself exactly after an integer multiple of a full turn, so $G_P = 2\pi\mathbb{Z}$ as a subgroup of $(\mathbb{R}, +)$. In particular, this real-number action is not faithful: 2π acts trivially.

Exercise 16.14: Unions of disjoint actions

(a) If the X_i are disjoint, every x in their union belongs to exactly one X_i. Define gx using that component's given action. This is well defined, stays in X_i, and satisfies $ex = x$ and $(gh)x = g(hx)$ because the component action does.

(b) Every $x \in X$ belongs to Gx. Two orbits meeting at a point are equal: if $gx = hy$, then $x = g^{-1}hy$, so every image of x is an image of y, and conversely. Thus the distinct orbits partition X, and their union, with the component actions, is exactly the original action.

Exercise 16.15: A transitive action is a coset action

Put $H = G_{x_0}$ and define $\phi(gx_0) = gH$; transitivity provides such a g for each point.

Prove the definition does not depend on the choice of g.

If $gx_0 = hx_0$, then $(h^{-1}g)x_0 = x_0$, so $h^{-1}g \in H$ and $gH = hH$. The same equivalences in reverse show that $gH = hH$ implies $gx_0 = hx_0$. Thus ϕ is both well defined and injective.

Check surjectivity and the action.

Every coset gH is the image of gx_0, so ϕ is onto. For $k \in G$,

$$\phi(k(gx_0)) = \phi((kg)x_0) = (kg)H = k\phi(gx_0).$$

This proves the required G-set isomorphism.

Exercise 16.16: Decomposing an arbitrary action

First partition X into its distinct orbits O_i. Choose a point x_i in each nonempty orbit and let $H_i = G_{x_i}$. Exercise 16.15 supplies an isomorphism $O_i \to G/H_i$, where G/H_i here denotes the *set of left cosets*; H_i need not be normal. Tag the coset set by i and let

$$g \cdot (kH_i, i) = (gkH_i, i).$$

The map taking $x \in O_i$ to $(\phi_i(x), i)$ is bijective, because the orbits partition X and each ϕ_i is bijective. It is equivariant in each component and hence globally. The tags prevent cosets from different components from being confused. For an arbitrary infinite family, choosing all the representatives uses the usual choice convention; finite actions need no such qualification.

Exercise 16.17: When two coset actions are isomorphic

(a) Let $H = G_{x_0}$. An element k fixes g_0x_0 exactly when $(g_0^{-1}kg_0)x_0 = x_0$, equivalently $g_0^{-1}kg_0 \in H$. Consequently $G_{g_0x_0} = g_0Hg_0^{-1}$.

(b) The conjecture is that G/H and G/K are isomorphic as G-sets exactly when H and K are conjugate subgroups.

(c) If $K = tHt^{-1}$, define $\phi(gH) = gt^{-1}K$. Replacing g by gh for $h \in H$ does not change the answer, because $ght^{-1}K = gt^{-1}(tht^{-1})K = gt^{-1}K$. Its inverse sends uK to utH, and left multiplication commutes with ϕ, proving sufficiency. Conversely, an equivariant bijection sends H to some aK and preserves stabilizers. The stabilizer of H is H, while that of aK is aKa^{-1}. Thus $H = aKa^{-1}$, proving necessity.

Exercise 16.18: All transitive actions of the group of order four

Plan. A transitive action is a coset action, and isomorphism types correspond to conjugacy classes of subgroups. Since $\mathbb{Z}_4$ is abelian, conjugation does not identify different subgroups. Its subgroups are $\mathbb{Z}_4$, $\{0, 2\}$, and $\{0\}$, giving exactly three types, with $1, 2, 4$ points respectively. Their action tables are

	a
0	a
1	a
2	a
3	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a

	a	b	c	d
0	a	b	c	d
1	b	c	d	a
2	c	d	a	b
3	d	a	b	c

.

In the four-point table, a, b, c, d represent $0, 1, 2, 3$; in the two-point table they represent the two cosets of $\{0, 2\}$.

Exercise 16.19: All transitive actions of the group of order six

There is one subgroup of each order $1, 2, 3, 6$ in $\mathbb{Z}_6$ and no others, since every subgroup of a cyclic group is cyclic. Conjugation is trivial. The four isomorphism types have sizes $6, 3, 2, 1$. For the first

three smaller sizes the tables are

	a
0	a
1	a
2	a
3	a
4	a
5	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a
4	a	b
5	b	a

	a	b	c
0	a	b	c
1	b	c	a
2	c	a	b
3	a	b	c
4	b	c	a
5	c	a	b

.

The six-point regular action is

	a	b	c	d	e	f
0	a	b	c	d	e	f
1	b	c	d	e	f	a
2	c	d	e	f	a	b
3	d	e	f	a	b	c
4	e	f	a	b	c	d
5	f	a	b	c	d	e

.

Each row adds its group element to the residue represented by the column, modulo the number of points. Since that number divides 6, this defines an action of $\mathbb{Z}_6$ in every case.

Exercise 16.20: All transitive actions of the symmetric group on three letters

Write $r = (1,2,3)$, $r^2 = (1,3,2)$, $s = (2,3)$, $t = (1,3)$, and $u = (1,2)$. The subgroups are S_3, A_3, the three conjugate order-two subgroups, and the identity subgroup. Hence there are four transitive isomorphism types, with $1, 2, 3, 6$ points. The one-point action is trivial; the two-point action records even or odd parity; the three-point action is the natural permutation action; and the six-point action is left multiplication on S_3. Here are all their action tables, with the requested row order:

	a
e	a
r	a
r^2	a
s	a
t	a
u	a

	a	b
e	a	b
r	a	b
r^2	a	b
s	b	a
t	b	a
u	b	a

	a	b	c
e	a	b	c
r	b	c	a
r^2	c	a	b
s	a	c	b
t	c	b	a
u	b	a	c

,

	a	b	c	d	e	f
e	a	b	c	d	e	f
r	b	c	a	f	d	e
r^2	c	a	b	e	f	d
s	d	e	f	a	b	c
t	e	f	d	c	a	b
u	f	d	e	b	c	a

.

In the last table the point labels a, b, c, d, e, f represent the group elements e, r, r^2, s, t, u respectively. For example, $rs = u$, so row r, column d has entry f. The point label e in that table is distinct from the row label for the group identity.